been trained on a corpus with a corresponding political bias, positioning them at opposite locations in the ideological spectrum, as suggested by their names [5], [11].

In summary, this report provides a comprehensive and multifaceted analysis of political bias in AI systems by employing four complementary methodologies and integrating them into a combined ranking of political bias in AI systems. This approach not only addresses the limitations of previous studies on AI ideological bias but also offers new insights into the nature and extent of political bias in AI-generated content.

Our analysis has been done from mostly an American standpoint and results reported herein might not be applicable to other regions of the world. Through this report, the aim is to contribute to the responsible development and deployment of AI technologies by highlighting the importance of detecting and mitigating political biases in AI systems used by millions of users.

Measuring political bias in AI systems using multiple methodologies

Comparing AI-generated text with the language used by U.S. Congress legislators

A 2010 study measured U.S. media bias by comparing language used by news media outlets with that used by Democratic and Republican U.S. legislators [10]. It found that left-leaning news outlets tended to use expressions that were commonly used by Democrats (i.e. Iraq war, state tax, etc.), while right-leaning outlets tended to use language favored by Republicans (i.e. war on terror, death tax, etc.). This suggests a clear alignment between linguistic choices and political leanings.

I use a similar methodology to determine if language generated by LLMs is more akin to terms commonly associated with Democratic or Republican members of the U.S. Congress in their congressional remarks. To do that, I derive two 1,000 two-word terms (i.e. bigrams) sets with high partisan contrast (highly used by representatives from one party and comparatively less used by representatives from the other party in U.S. congressional remarks) (see the Methods section for details). Figure 1 shows the results of that analysis by

displaying terms highly used by members of congress from each party in relation to their counterparts from the other party. As the figure makes clear, Democrats members disproportionately refer in their remarks to *affordable care*, *gun violence*, *African Americans*, *domestic violence*, *minimum wage,* or *voting rights*; while Republicans disproportionately emphasize *balanced budgets*, the *southern border*, *illegal immigrants*, *religious freedom*, *job creators, tax increases, government spending* or *national defense*.

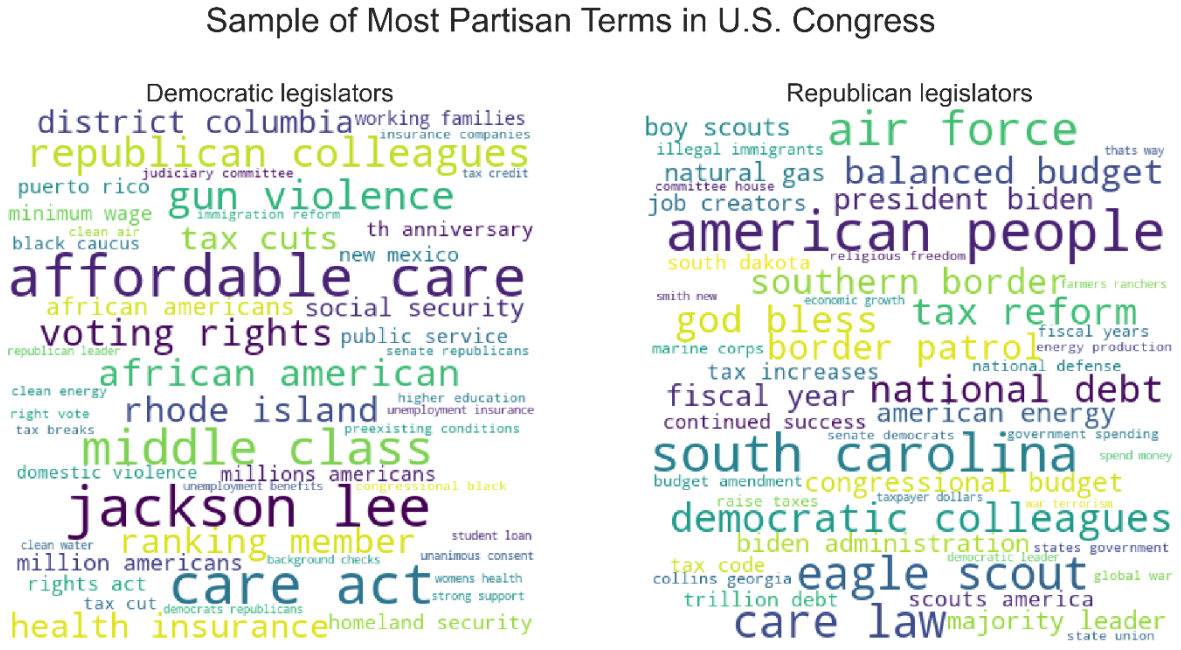

Figure 1 Bigrams with high contrast partisan usage: The left subplot highlights bigrams predominantly used by Democratic Congress members in the Congressional Record compared to their Republican counterparts. The right subplot displays the opposite trend, showcasing bigrams more frequently employed by Republican Congress members relative to Democrats.

I then asked the studied LLMs to generate thousands of policy recommendations about a variety of topics such as Social Security, Medicare and Medicaid, criminal justice reform, voting rights, immigration, states' rights, or second amendment rights. I also prompted the target LLMs to generate commentary about politically aligned public figures (U.S. Presidents, Senators, Governors, Supreme Court Justices, journalists, and political leaders of Western countries). I then measured the difference in the usage of the two sets of partisan bigrams between the LLMs-generated text snippets and Republican/Democratic usage of the same terms. Figure 2 displays the difference in Jensen-Shannon Divergence (JSD) between the distribution of partisan terms frequencies in LLM-generated text and the distribution of those terms in U.S. congressional remarks by Republicans and Democrats, respectively. Basically, negative values

in the x-axis indicate a closer alignment between LLMs outputs and language patterns used by Democratic legislators, while positive values indicate a closer alignment between LLMs outputs and language patterns used by Republican legislators.

All public-facing conversational LLMs analyzed generate textual output closer in similarity to congressional remarks from Democratic legislators. Base LLMs show the same directionality as conversational models, but the skew is much milder. Ideologically aligned LLMs (LeftwingGPT and RightwingGPT) exhibit predictable patterns: LeftwingGPT uses significantly more Democratic bigrams, while RightwingGPT uses more Republican bigrams, though only slightly.

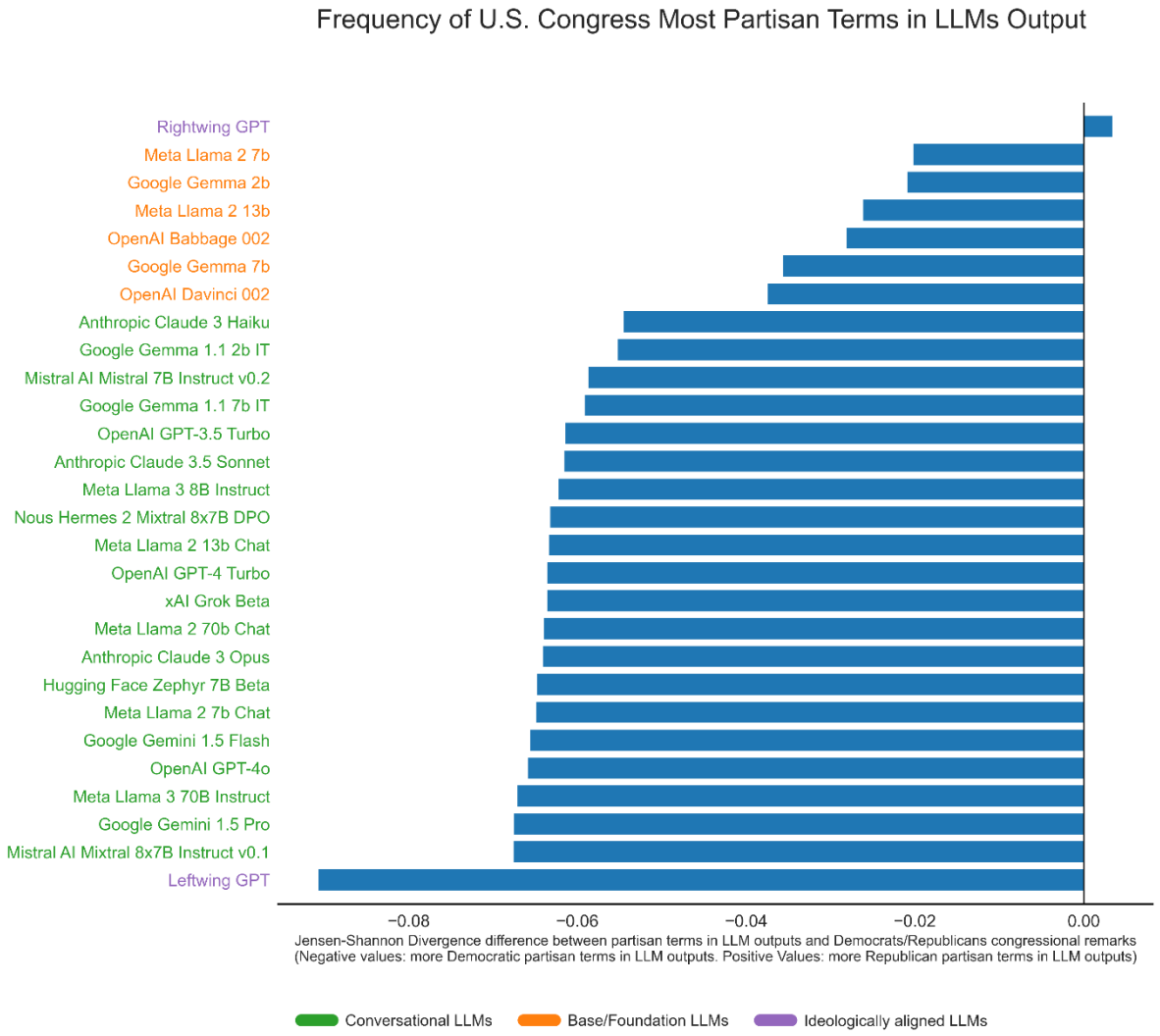

Figure 2 LLMs usage of partisan bigrams preferentially used by Democrats in U.S. Congressional record (negative values) and partisan bigrams preferentially used by Republicans (positive values).

Notably, the asymmetry in partisan term usage is more marked for LeftwingGPT than it is for RightwingGPT. Perhaps LeftwingGPT is simply more ideologically skewed, but it is also possible that Republican members of